

- **Solving intractable problems:** Quantum computing holds promise in solving problems that are currently deemed computationally intractable. Tasks such as factoring large numbers, optimising complex systems, and simulating quantum physics phenomena become more feasible with quantum algorithms, paving the way for advancements in cryptography, optimisation, and scientific research.
- **Revolutionising artificial intelligence:** The exponential speedup offered by quantum computing has the potential to revolutionise artificial intelligence (AI). Quantum machine learning algorithms could dramatically enhance pattern recognition, optimisation processes, and data analysis, leading to significant advancements in AI applications.
- **Optimising supply chains and financial models:** Quantum computing's ability to process massive data sets in real-time could revolutionise industries like finance and logistics. Complex optimisation problems involved in supply chain management, portfolio optimisation, and risk assessment can be tackled more efficiently, fostering innovation, and economic growth.
- **Drug discovery and material science:** The quantum nature of particles allows quantum computers to simulate molecular and atomic interactions with unprecedented accuracy. This capability is a game-changer in drug discovery, material science, and chemistry, potentially accelerating the development of new pharmaceuticals and advanced materials.

Evolution of quantum computing technology

- **Conceptual foundations:** Quantum computing is rooted in the foundational principles of quantum mechanics, with visionaries like

Richard Feynman and David Deutsch laying the conceptual groundwork in the 1980s. Their insights paved the way for a new approach to computation that harnessed the unique properties of quantum systems.

- **The birth of quantum algorithms:** The development of quantum algorithms, notably Shor's algorithm for factoring large numbers and Grover's algorithm for searching an unsorted database, marked a significant milestone in the 1990s. These algorithms showcased the potential advantages of quantum computation over classical methods.
- **Rise of quantum hardware:** The past decade witnessed remarkable progress in quantum hardware development. Quantum processors, based on technologies like superconducting qubits, trapped ions, and topological qubits, are becoming increasingly sophisticated and reliable. A global race is on to achieve quantum supremacy, where a quantum computer outperforms the most advanced classical supercomputers.
- **Quantum cloud computing:** Quantum computing is transitioning from academic research to practical applications through cloud-based services. Companies offer access to quantum processors and simulators via the cloud, democratising quantum computing for researchers and businesses to experiment without extensive hardware resources.

Quantum development frameworks and simulation tools

Quantum development frameworks and simulation tools play a pivotal role in quantum computing, providing essential resources for researchers and developers to explore and harness the unprecedented capabilities of quantum systems. These

tools are the backbone of quantum programming, offering platforms for designing, simulating, and optimising quantum algorithms before deployment on actual quantum processors. These frameworks and tools, which include IBM's Qiskit and Google's Cirq, not only propel quantum algorithm development but also contribute to the collaborative and dynamic landscape of quantum research and innovation.

IBM quantum development kit - Qiskit

Qiskit stands at the forefront of quantum development frameworks, spearheaded by IBM as an open source initiative. It provides a robust and comprehensive toolkit for quantum computing, offering a wide array of features that cater to both novices and seasoned quantum developers.

Key components

Qiskit Terra: At the heart of Qiskit is Terra, the foundational component for quantum circuit design and optimisation. Terra allows users to define and manipulate quantum circuits with ease, enabling the creation of complex algorithms through a straightforward and intuitive interface.

Qiskit Aer: Qiskit Aer is a high-performance simulator designed for accurate quantum circuit simulations. This component is instrumental during the development phase, allowing developers to test and debug quantum algorithms before deploying them on actual quantum hardware. Aer supports a variety of noise models, enhancing the fidelity of simulations.

Qiskit Ignis: Addressing the challenges of noisy quantum processors, Qiskit Ignis provides tools for characterising and mitigating errors in quantum systems. Ignis enables developers to optimise the performance of quantum algorithms in the presence of noise, contributing to the advancement of practical quantum computing.